

Paper Validation Report

for DiMag



Correspondence percentages: 17%

Number of experiments found: 11

Experiment 1. Formulation stated: "first, we manually collected 1282 raw samples from 126 unique scientific articles (Table S1, Supporting Information) with descriptions of synthesis, materials characterizations, and measurements of magnetic hyperthermia and MRI performances in vitro."

Implementation found: ['NOT FOUND'] **Missing components:** ['Automated or manual data-extraction routines for sourcing 1282 samples from 126 articles', 'Code that builds Table S1 or any structured dataset from raw literature descriptions', 'Parsing/curation pipeline converting synthesis and characterization text into machine-readable format'] **Correspondence:** 5.0%

Experiment 2. Formulation stated: "Average magnetic moment of metals in the core was calculated using magnetic moments of each element with help of stoichiometry to better describe composition of nanoparticles (Equation (S1), Supporting Information)." **Implementation found:** ['NOT FOUND'] **Missing components:** ['Computation of average magnetic moment of core metals using stoichiometry (Equation S1)', 'Lookup or storage of elemental magnetic moment values', 'Integration of magnetic moment calculation into nanoparticle descriptor pipeline']

Correspondence: 5.0%

Experiment 3. Formulation stated: "Average spin of metal elements on the surface was obtained to describe interactions between the surface elements and proton spins responsible for relaxation properties of the nanoparticle (Equation (S2), Supporting Information)." **Implementation found:** ['NOT IMPLEMENTED'] **Missing components:** ['Computation of average spin of surface metal elements (Equation S2)', 'Simulation or calculation of interactions between surface metal spins and proton spins', 'Integration of spin-derived features into dataset generation or model input pipeline', 'Any reference to proton relaxation mechanisms in feature engineering or modeling']

Correspondence: 5.0%

Experiment 4. Formulation stated: "Volume and surface area were calculated using the shape and size of the nanoparticle to describe morphological characteristics of nanomagnets."

Implementation found: ['NOT FOUND'] **Missing components:** ['Volume calculation from nanoparticle shape and size', 'Surface area calculation from nanoparticle shape and size', 'Any function/class that derives morphological descriptors from raw dimensions', 'Integration of computed volume and surface area into the modelling dataset'] **Correspondence:** 5.0%

Experiment 5. Formulation stated: "We calculated and analyzed the Pearson correlation matrix (Figure 2A)." **Implementation found:** ['NOT FOUND'] **Missing components:** ['Pearson correlation matrix calculation', 'Visualization of the correlation matrix (Figure 2A)'] **Correspondence:** 5.0%

Experiment 6. Formulation stated: "we chose kNN imputation technique, because it considers the relationships between the different features, it is robust to outliers, and it can handle both categorical and continuous variables.[42]" **Implementation found:** ['NOT FOUND'] **Missing components:** ['kNN imputation step for handling missing data across categorical and continuous features'] **Correspondence:** 0.0%

Experiment 7. Formulation stated: "To cope with that, we first removed the total of 211 samples such that each sample in the remaining dataset had no more than 15% of missing values."

Implementation found: ['NOT FOUND'] **Missing components:** ['Removal of 211 samples exceeding 15% missing values (no row-level missing-data filtering present in any script)']

Correspondence: 5.0%

Experiment 8. Formulation stated: "After that, we applied the kNN imputation algorithm to fill the gaps and obtain the complete dataset." **Implementation found:** ['NOT FOUND'] **Missing components:** ['kNN imputation step to fill missing values'] **Correspondence:** 5.0%

Experiment 9. Formulation stated: "We normalized and log-scaled the data for each problem before splitting to train and test sets in the 80/20 ratio." **Implementation found:** ['attached_file.py:', 'provided_script.py:', 'r2_modeling.py:', 'r1_modeling.py:', 'source.py:', 'script.py:', 'model_training_script.py:', 'r1_extra_trees_regression.py:', 'r2_relaxivity_et_regression.py:']

Missing components: ['Normalization and log-scaling performed BEFORE the 80/20 split']

(currently done after splitting in all scripts)', 'Log-scaling of input feature columns (only the target column is log-transformed)'] **Correspondence:** 75.0%

Experiment 10. Formulation stated: "hyperparameter tuning was done using the model-agnostic Optuna framework[46] (Table S5, Supporting Information)." **Implementation found:** ['NOT FOUND'] **Missing components:** ['Optuna-based hyperparameter tuning routine (no optuna import or study/optimization found in any script)'] **Correspondence:** 0.0%

Experiment 11. Formulation stated: "For each regression problem, we optimized and compared performance of four models (Figure S4, Supporting Information)." **Implementation found:** ['attached_file.py:lines 1-400 (builds and evaluates ExtraTrees, RandomForest, LightGBM, XGBoost for SAR)', 'provided_script.py:lines 1-380 (builds and evaluates ExtraTrees, RandomForest, LightGBM, XGBoost for r1)', 'r2_modeling.py:lines 1-360 (builds and evaluates ExtraTrees, RandomForest, LightGBM, XGBoost for r2)'] **Missing components:** ['Automated hyper-parameter optimisation routine (only fixed, pre-tuned parameters present)', 'Unified aggregation/visualisation of cross-model comparisons across all targets (Figure S4 generation not implemented)', 'Statistical significance testing between model performances (not found)'] **Correspondence:** 80.0%